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Problem:

Given $q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1})$

Show $q(x_{1:T} | x_0) = q(x_T | x_0) \prod_{t=1}^{T-1} q(x_{t+1} | x_t, x_0)$

Hint:

$$q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1}) = \prod_{t=1}^T q(x_t | x_{t-1}, x_0)$$

$$= q(x_1 | x_0) \cdot \prod_{t=2}^T \frac{q(x_t, x_{t-1}, x_0)}{q(x_{t-1}, x_0)}$$

$$= q(x_1 | x_0) \cdot \prod_{t=2}^T \frac{q(x_0) q(x_t, x_{t-1} | x_0)}{q(x_0) q(x_{t-1} | x_0)}$$

$$= q(x_1 | x_0) \cdot \prod_{t=2}^T \frac{q(x_t | x_0) q(x_{t-1} | x_t, x_0)}{q(x_{t-1} | x_0)}$$

$$= q(x_1 | x_0) \cdot \prod_{t=2}^T \frac{q(x_t | x_0)}{q(x_{t-1} | x_0)} \cdot \prod_{t=2}^T q(x_{t-1} | x_t, x_0)$$

↓ 約分

$$= q(x_1 | x_0) \cdot \frac{q(x_T | x_0)}{q(x_1 | x_0)} \cdot \prod_{t=2}^T q(x_{t-1} | x_t, x_0)$$

$$= q(x_T | x_0) \cdot \prod_{t=2}^T q(x_{t-1} | x_t, x_0) \quad \#$$

$$\frac{\cancel{q(x_2 | x_0)} \cancel{q(x_3 | x_0)} \dots q(x_T | x_0)}{q(x_1 | x_0) \cancel{q(x_2 | x_0)} \dots \cancel{q(x_{T-1} | x_0)}}$$

$$= \frac{q(x_T | x_0)}{q(x_1 | x_0)}$$

Prove: $q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\alpha_t} x_0, (1 - \alpha_t) I)$ Q2 (4)

$$x_t \sim q(x_t | x_0).$$

$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1 - \alpha_t} \epsilon_{t-1} \quad ; \quad \text{where, } \epsilon_{t-1}, \epsilon_{t-2}, \dots \sim \mathcal{N}(0, I)$$

$$= \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} x_{t-2} + \sqrt{1 - \alpha_{t-1}} \epsilon_{t-2}^*) + \sqrt{1 - \alpha_t} \epsilon_{t-1}^*$$

$$= \sqrt{\alpha_t \alpha_{t-1}} \cdot x_{t-2} + \sqrt{\alpha_t - \alpha_t \alpha_{t-1}} \epsilon_{t-2}^* + \sqrt{1 - \alpha_t} \epsilon_{t-1}^*$$

$$= \sqrt{\alpha_t \alpha_{t-1}} \cdot x_{t-2} + \sqrt{\alpha_t - \alpha_t \alpha_{t-1} + 1 - \alpha_t} \epsilon_{t-2}$$

$$= \sqrt{\alpha_t \alpha_{t-1}} \cdot x_{t-2} + \sqrt{\alpha_t - \alpha_t \alpha_{t-1} + 1 - \alpha_t} \epsilon_{t-2}$$

$$= \sqrt{\alpha_t \alpha_{t-1}} \cdot x_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \cdot \epsilon_{t-2}, \text{ where } \epsilon_{t-2} \text{ merge two Gaussian.}$$

$$= \sqrt{\prod_{i=1}^t \alpha_i} x_0 + \sqrt{1 - \prod_{i=1}^t \alpha_i} \epsilon_0$$

$$= \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon_0 \sim \mathcal{N}(x_t; \sqrt{\alpha_t} x_0, \sqrt{1 - \alpha_t}^2 \epsilon_0)$$

$$\Rightarrow \mathcal{N}(x_t; \sqrt{\alpha_t} x_0, (1 - \alpha_t) \epsilon_0)$$

Prove $g(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I)$ Q2(b).

where $\tilde{\mu}_t(x_t, x_0) := \frac{\sqrt{\alpha_t} \beta_t}{1-\alpha_t} x_0 + \frac{\sqrt{\alpha_t}(1-\alpha_{t-1})}{1-\alpha_t} x_t$, $\tilde{\beta}_t := \frac{1-\alpha_{t-1}}{1-\alpha_t} \beta_t$.

$$g(x_{t-1}|x_t, x_0) = g(x_t|x_{t-1}, x_0) \frac{g(x_{t-1}|x_0)}{g(x_t|x_0)}$$

$$\propto \exp\left(-\frac{1}{2}\left(\frac{(x_t - \sqrt{\alpha_t} x_0)^2}{\beta_t} + \frac{(x_{t-1} - \sqrt{\alpha_{t-1}} x_0)^2}{1-\alpha_{t-1}} - \frac{(x_t - \sqrt{\alpha_t} x_0)^2}{1-\alpha_t}\right)\right)$$

$$= \exp\left(-\frac{1}{2}\left(\frac{x_t^2 - 2\sqrt{\alpha_t} x_t x_0 + \alpha_t x_0^2}{\beta_t} + \frac{x_{t-1}^2 - 2x_{t-1}\sqrt{\alpha_{t-1}} x_0 + \alpha_{t-1} x_0^2}{1-\alpha_{t-1}} - \frac{(x_t - \sqrt{\alpha_t} x_0)^2}{1-\alpha_t}\right)\right)$$

$$= \exp\left(-\frac{1}{2}\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1-\alpha_{t-1}}\right)x_{t-1}^2 - \left(\frac{2\sqrt{\alpha_t}}{\beta_t} x_t + \frac{2\sqrt{\alpha_{t-1}}}{1-\alpha_{t-1}} x_0\right)x_{t-1} + C\right)$$

$$\tilde{\beta}_t = \left(\frac{\alpha_t}{\beta_t} + \frac{1}{1-\alpha_{t-1}}\right)^{-1} = \left(\frac{\alpha_t - \alpha_t + \beta_t}{\beta_t(1-\alpha_{t-1})}\right)^{-1} = \left(\frac{1-\alpha_t}{\beta_t(1-\alpha_{t-1})}\right)^{-1} = \frac{1-\alpha_{t-1}}{1-\alpha_t} \cdot \beta_t$$

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$$\tilde{\mu}_t(x_t, x_0) = \left(\frac{\sqrt{\alpha_t}}{\beta_t} x_t + \frac{\sqrt{\alpha_{t-1}}}{1-\alpha_{t-1}} x_0\right) / \left(\frac{\alpha_t}{\beta_t} + \frac{1}{1-\alpha_{t-1}}\right)$$

$$= \left(\frac{\sqrt{\alpha_t}}{\beta_t} x_t + \frac{\sqrt{\alpha_{t-1}}}{1-\alpha_{t-1}} x_0\right) \frac{1-\alpha_{t-1}}{1-\alpha_t} \cdot \beta_t$$

$$= \frac{\sqrt{\alpha_t}(1-\alpha_{t-1})}{1-\alpha_t} x_t + \frac{\sqrt{\alpha_{t-1}}\beta_t}{1-\alpha_t} x_0$$

Prove $L_{t+1} = \mathbb{E}_q \left[\frac{1}{2\sigma^2} \|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2 \right] + C \quad Q2(8)$

$$\arg\min_{\theta} D_{KL}(q(x_{t+1} | x_t, x_0) \| p_\theta(x_{t+1} | x_t))$$

$$= \arg\min_{\theta} D_{KL}(N(x_{t+1}; \mu_g, \Sigma_g(t)) \| N(x_{t+1}; \mu_\theta, \Sigma_g(t)))$$

$$= \arg\min_{\theta} \frac{1}{2} \left[\log \frac{|\Sigma_g(t)|}{|\Sigma_g(t)|} - d + \text{tr}(\Sigma_g(t)^T \Sigma_g(t)) + (\mu_\theta - \mu_g)^T \Sigma_g(t)^{-1} (\mu_\theta - \mu_g) \right]$$

$$= \arg\min_{\theta} \frac{1}{2} [1 - d + d + (\mu_\theta - \mu_g)^T \Sigma_g(t)^{-1} (\mu_\theta - \mu_g)]$$

$$= \arg\min_{\theta} \frac{1}{2} [(\mu_\theta - \mu_g)^T \Sigma_g(t)^{-1} (\mu_\theta - \mu_g)]$$

$$= \arg\min_{\theta} \frac{1}{2} [(\mu_\theta - \mu_g)^T (\sigma_g^2(t) I)^{-1} (\mu_\theta - \mu_g)]$$

$$= \arg\min_{\theta} \frac{1}{2\sigma_g^2(t)} [\|\mu_\theta - \mu_g\|^2] \sim \mathbb{E} \left[\frac{1}{2\sigma^2} \|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2 \right]$$

∴ optimizing the KL divergence term reduces to minimizing the difference between the means of the two distributions.